

Object-Centric Causal Discovery in MIMIC-IV
A discovery-to-inference workflow for identifying candidate upstream drivers of hospital-acquired pressure injury

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Abstract

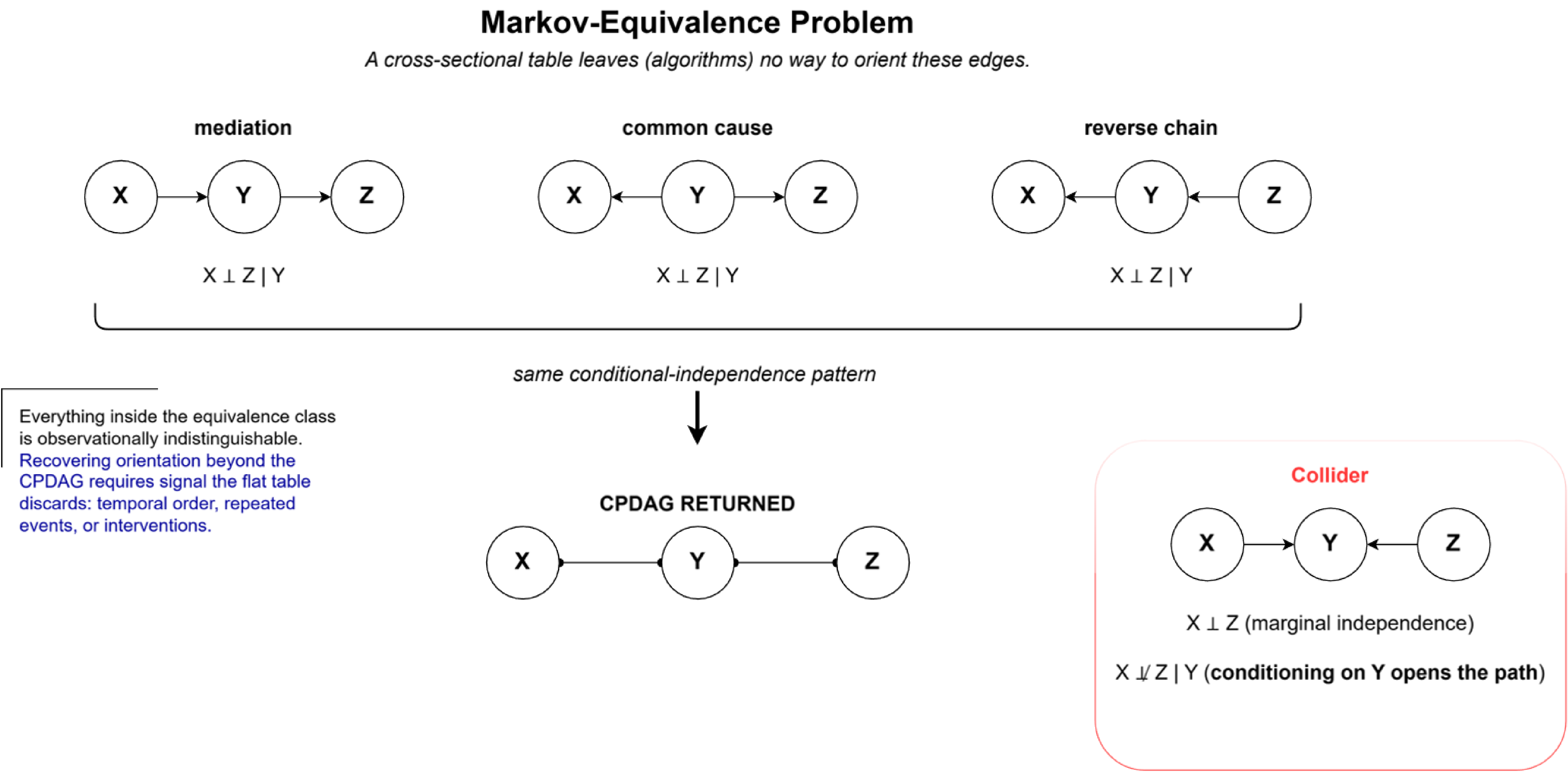
BACKGROUND / OBJECTIVE

Traditional causal discovery algorithms face critical limitations in healthcare: they assume “flat” datasets, struggle with EHR relational complexity, and yield equivalence classes rather than fully-oriented graphs. We present an object-centric approach that encodes clinical constraints (temporal precedence, relational structure, informative missingness) while enabling data-driven discovery within that structure. We applied this method to identify candidate causal factors for hospital-acquired pressure injuries (HAPrI) in MIMIC-IV (v3.1).

METHODS

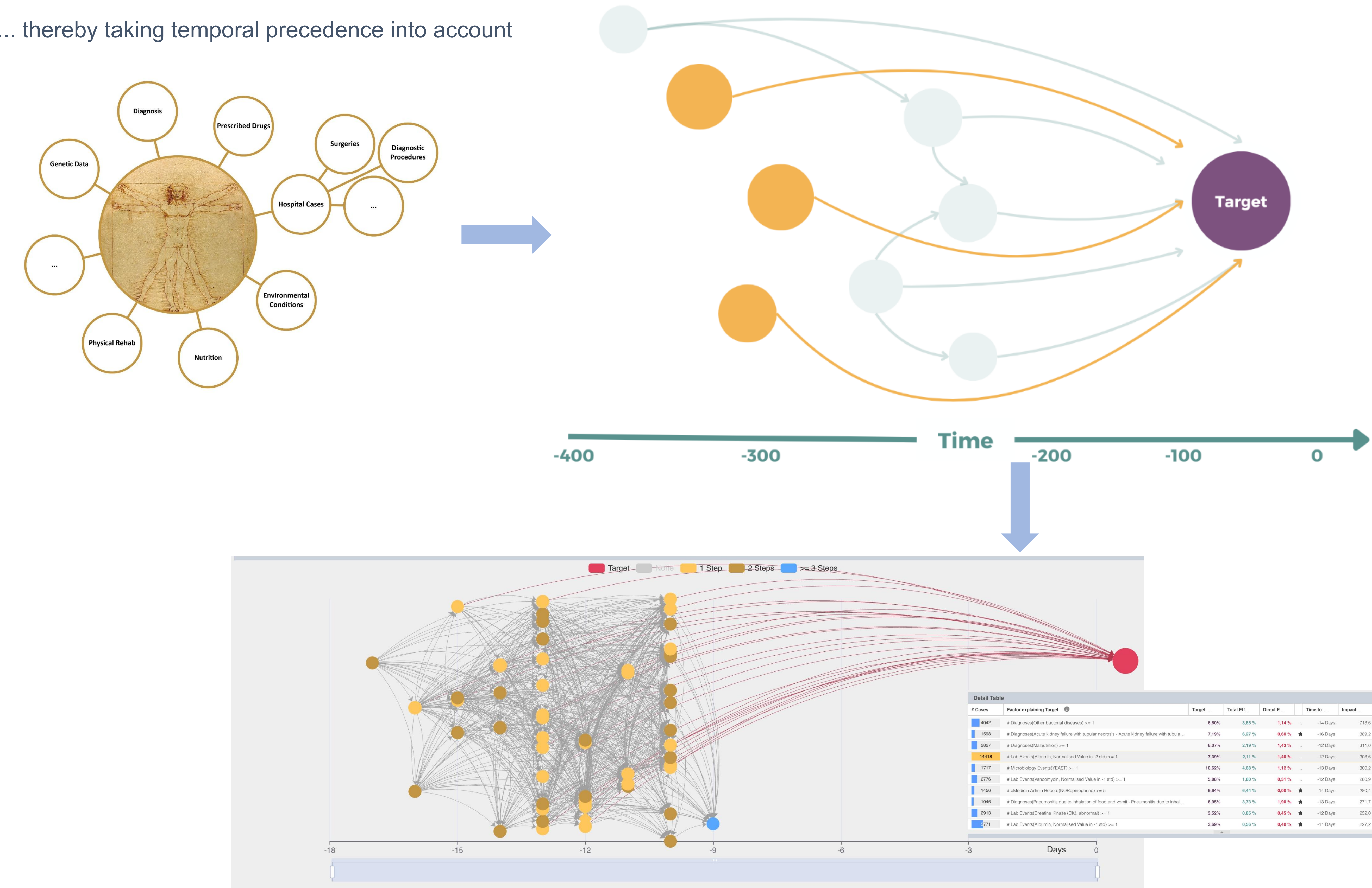
Using MIMIC-IV adult ICU admissions, we represented patient journeys as temporally ordered “objects” (admissions, procedures, medications, labs) and applied a causal discovery algorithm operating directly on this relational structure without flattening. Domain knowledge constrained the search space, enforcing temporal precedence to minimize target leakage and encoding missingness mechanisms, while the algorithm screened millions of candidate dependencies to construct a sparse causal graph.

Classic Causal Discovery on a flat table returns an equivalence class, not an oriented DAG



The causal-discovery search operates directly on this relational/object structure

... thereby taking temporal precedence into account



Results from a real-world data example:

Hospital-acquired pressure injuries in the ICU: Using MIMIC IV v 3.1:

RESULTS

The algorithm prioritized hypoalbuminemia as a *candidate upstream driver* of HAPrI, pending clinician review and confirmatory target-trial / TMLE analysis. It rediscovered established risk factors (mechanical ventilation, hemodynamic instability, immobility, moisture exposure, yeast colonization) without manual feature engineering, and flagged fluid management patterns and albumin administration timing as secondary candidate edges.

CONCLUSIONS

Object-centric causal discovery offers a middle path between purely data-driven search and expert-specified DAGs: clinical knowledge disciplines the algorithm, while the algorithm surfaces relationships experts might overlook. Future work will engage clinicians to refine the graph and employ modern causal estimators (TMLE, longitudinal g-methods) to quantify effects of albumin and other interventions on HAPrI risk.

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